

Tharun Marella

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Build and ship Python/FastAPI backends for AI/ML products: agentic, retrieval, and voice systems with real evaluation, deployed to production on AWS EKS. 4+ years operating backend systems at scale, with AI platforms built from scratch (Recall, Tellora, Prism).

TECHNICAL SKILLS

Backend & APIs: Python (async/asyncio), Java, FastAPI, Pydantic, SQLAlchemy, Celery/ARQ, Spring Boot, Node.js, GraphQL, REST, WebSocket/SSE streaming, OAuth2/JWT, pytest

AI / LLM: Agentic systems (LangGraph), RAG & hybrid retrieval, reranking, eval harnesses & LLM observability (LangSmith, Helicone), fine-tuning (QLoRA), LLM serving (vLLM), multi-provider orchestration (LiteLLM), vector search

Frontend: React, Next.js, TypeScript, JavaScript

Data, Infra & Ops: PostgreSQL/pgvector, Redis, MongoDB, Qdrant, Meilisearch, S3/R2, AWS (EKS), Docker, Kubernetes, Kafka, RabbitMQ, GitHub Actions/Jenkins (CI/CD), Sentry, structured logging

PROFESSIONAL EXPERIENCE

Software Development Engineer

McLean, VA (Remote)

Freddie Mac

Aug 2022 – Present

- Lead end-to-end ownership of a seller photo appraisal pipeline where vision LLMs score property condition and annotate defects from submitted home images on FastAPI, replacing a manual-only review workflow with structured, automated condition reports.
- Raised FredAI Chat retrieval accuracy 50% and held p95 latency under 100ms by building a hybrid semantic + lexical search platform over the Seller/Service Guide with cross-encoder reranking and an offline eval harness that A/B-tested precision and recall.
- Shipped FredAI's production chat backend: SSE-streamed responses, custom GPT and prompt-suggestion APIs, index-management admin, and LiteLLM failover on provider rate limits and saturation.
- Built async Seller/Service Guide ingestion (Celery, Pydantic) that chunked, embedded, and re-indexed policy content on publish, grounding answers in citation-backed sources for Guide refactoring workflows.
- Scaled FredAI from MVP pilot to 50K+ internal users through 5x traffic spikes with zero dropped requests by deploying Dockerized services to AWS EKS with predictive autoscaling and LangSmith trace-based latency analysis across model providers.

Software Development Engineer

San Mateo, CA

Luma Health

May 2021 – July 2022

- Built Spring Boot microservices secured with Spring Security (OAuth2/JWT) for scheduling, patient messaging, and bidirectional EHR sync (Epic, athenahealth) on Luma's patient-engagement platform, shipped through Jenkins CI/CD pipelines.
- Cut a nightly patient-data reconciliation from 6h to 45 min by re-architecting a Spring Batch pipeline that chunked and parallelized Spring Data MongoDB reads/writes across appointment and provider collections, with JUnit/Mockito tests for chunk processors and repository layers.
- Scaled COVID vaccination scheduling throughput 3x by optimizing MongoDB aggregation pipelines and Spring Data custom repositories for high-volume waitlist and slot-availability lookups.
- Cut incident-detection time 70% with Elasticsearch/Kibana alerting on core Java services.
- Decoupled appointment lifecycle, outbound messaging, and EHR-sync workflows with Kafka-backed Spring services and AWS Lambda triggers.

PROJECTS

Recall — *Voice AI backend: call capture, memory extraction, and search over telephony*

- Built production telephony pipeline: Twilio dual-channel Deepgram STT → Gemini structured extraction → ARQ workers that persist commitments, contact facts, and pre-call briefs.
- Shipped call-memory APIs with semantic search over transcript history, plus webhook idempotency and subscription billing hooks.
- Deployed API and worker on Railway with Sentry error tracking and Langfuse LLM tracing.

Tellora — *Agentic CRM backend: multi-tenant SaaS APIs, LangGraph agents, thin Next.js client*

- Architected revenue-ops API layer: telephony/SMS hooks, Gmail/Outlook sync, Apollo import, pipeline CRUD, and LangGraph agents that score leads and emit approve-before-act proposals.
- Built background workers for prospect enrichment (Jina + Gemini ICP scoring) and company-signal discovery, with Clerk multi-tenancy and Stripe billing.

Prism — *AI commerce data platform: ingestion, vision enrichment, and search API*

- Lifted product-classification accuracy from 68% to 87% by fine-tuning Qwen-VL (QLoRA) for vision-based category routing and serving inference with vLLM.
- Architected multi-service platform (prism-worker → prism-indexenrich → prism-api) at 100K+ products/day via distributed crawling, RabbitMQ orchestration, vision enrichment, dual-index search on PostgreSQL/TimescaleDB.

EDUCATION

Master of Science, Computer Science

California State University, Fresno | 2021